PORTFOLIO PROJECT · REAL DATA: TRANSPORT CANADA NATIONAL COLLISION DATABASE, 2019 AND 2020

When and where do Canadian roads turn deadly?

Start here: four headline numbers, what the data shows, and what I would do about it.

The question: When, where and under what conditions do fatal collisions concentrate, and who is most at risk, so an insurer's analytics team knows what to monitor and what to say about safe driving?

Reset filters

Collisions, 2019 and 2020

3,225 fatal, 182,547 injury

185,772

Share that are fatal

All-collision average is 1.74%

1.74%

Riskiest hour for a fatal outcome

At 4 am. Average is 1.74%

4.89%

Fatal collisions, 2020 vs 2019

All collisions -24.38%

-1.29%

What the data shows

- Collisions fell 24.38% in 2020, but fatal collisions only 1.29%, so the fatal share rose from 1.53% to 2.00%.
- At night (10 pm to 6 am) 3.41% of collisions are fatal against 1.52% by day. The riskiest hour is 4 am, at 4.89%.
- Head-on collisions are 10.96% fatal and cause 17.80% of fatal collisions from 2.82% of all. Occupants with no safety device used die 16.09 times as often (6.98% against 0.43%).

What I would do

- Monitor the fatal share, not only the count: in 2020 collisions fell but fatal ones did not.
- Put night driving, head-on and single-vehicle collisions, and roads with no traffic control on the risk watch list: they carry far more fatal outcomes than their share of collisions.
- Use the safety-device gap in safe-driving messaging. These are proposals from police-reported data, not claims or pricing advice.

How to use this report

- Use the page tabs. When shows the hour and weekday pattern, Compare sets 2020 against 2019, and Conditions is a what-if scenario: choose how much of the gap between risky and safer roads to close.
- Explore is a decomposition tree: click the + to split fatal collisions by any factor.
- Fatal share means the share of reported collisions that caused at least one death.

About the data and model

- Source: Transport Canada, National Collision Database, 2019 and 2020. Police-reported fatal and injury collisions, one row per person in each vehicle.
- A star schema: two fact tables (185,772 collisions and 463,630 people) and 15 dimensions. Every dimension has an 'unknown' member, so no record disappears from a chart.
- Collision counts here are not insurance claims.

Footnote: this is police-reported fatal and injury collisions only: no property-damage-only collisions and no claims, so it cannot support loss ratios or pricing. 2020 is a pandemic year, so it is not a typical comparison.

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Fatal collisions peak between midnight and 5 am

The fatal share of collisions by hour and weekday, with 2019 against 2020 by month.

Year (none = both)

All

Fatal collisions peak between midnight and 5 am: 3.41% at night against 1.52% by day.

Reset filters

Fatal share at night
Day (6 am to 10 pm) is 1.52%
3.41%

Fatal share by day
6 am to 10 pm, 163,444 collisions
1.52%

Night against day
Night is 2.24 times as likely to be fatal
2.24×

Riskiest single hour
At 4 am. Average is 1.74%
4.89%

Fatal share by hour and weekday

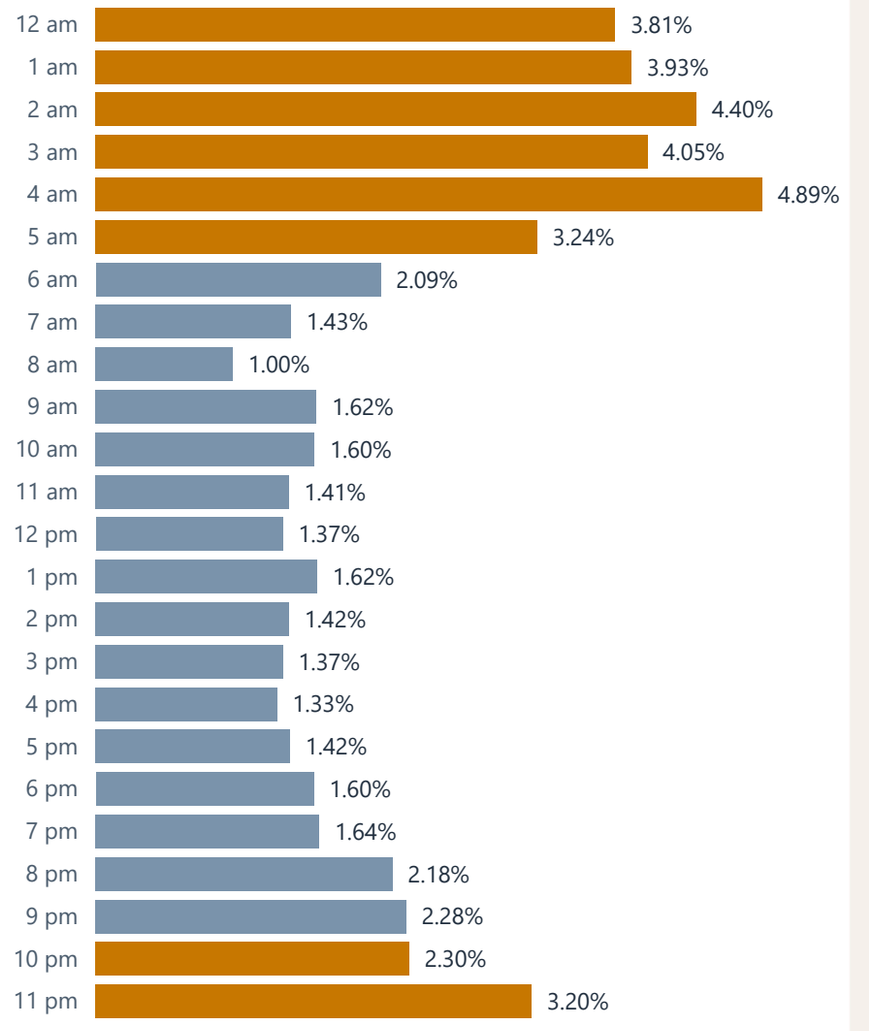
Darker means a higher fatal share. Cells with under 100 collisions are blank.

Hour	Mon	Tue	Wed	Thu	Fri	Sat	Sun
12 am	4.06%	4.51%	5.45%	2.08%	2.62%	4.59%	3.44%
1 am	3.27%	2.59%	3.27%	7.84%	4.03%	3.10%	4.22%
2 am	6.44%	2.53%	5.68%	3.21%	4.93%	3.93%	4.31%
3 am	3.03%	3.33%	2.17%	3.95%	2.62%	4.13%	6.32%
4 am	4.32%	5.19%	1.28%	5.49%	6.32%	6.79%	3.93%
5 am	4.71%	2.67%	2.38%	3.67%	3.09%	3.99%	2.11%
6 am	1.81%	1.81%	2.38%	1.99%	2.26%	2.17%	2.57%
7 am	1.47%	1.40%	1.46%	1.12%	1.31%	1.21%	3.22%
8 am	0.79%	1.02%	1.00%	0.78%	0.91%	1.54%	2.01%
9 am	2.01%	1.11%	1.26%	2.23%	1.49%	1.99%	1.27%
10 am	1.18%	1.48%	1.11%	1.16%	2.48%	1.28%	2.67%
11 am	1.53%	1.95%	1.42%	1.29%	1.20%	1.02%	1.58%
12 pm	1.17%	1.00%	1.04%	1.09%	1.77%	1.58%	1.93%
1 pm	1.36%	1.69%	1.99%	1.58%	1.25%	1.62%	1.91%
2 pm	1.46%	1.70%	1.73%	1.08%	1.24%	1.32%	1.45%
3 pm	1.26%	1.17%	1.34%	1.78%	1.11%	1.49%	1.52%
4 pm	0.95%	1.12%	1.27%	1.01%	1.49%	1.53%	2.41%
5 pm	1.24%	1.15%	1.12%	1.48%	1.15%	2.43%	1.79%
6 pm	1.58%	1.02%	1.50%	1.31%	1.87%	2.37%	1.66%
7 pm	1.03%	1.73%	1.26%	2.00%	2.03%	1.69%	1.58%
8 pm	2.37%	2.07%	1.96%	2.35%	1.85%	2.98%	1.65%
9 pm	2.08%	1.37%	2.24%	2.67%	3.05%	1.51%	2.95%
10 pm	3.21%	2.19%	2.54%	3.00%	2.04%	1.58%	1.78%
11 pm	2.52%	3.59%	3.37%	3.99%	3.79%	2.58%	2.49%

Times the overall average: under 0.75 | 0.75 to 1.5 | 1.5 to 2.5 | 2.5 to 4 | over 4

Fatal share by hour

Amber bars are the night hours, 10 pm to 6 am.



01 · Night against day

2.24×

Night collisions (10 pm to 6 am) are 3.41% fatal against 1.52% by day. The dark cells run from midnight to 5 am on every weekday.

02 · The riskiest hour

4.89%

At 4 am nearly 1 in 20 collisions is fatal, against the 1.74% average. The night bars are amber in the chart on the right.

03 · Read with care

1,400 vs 16,000

There are about 1,400 collisions at 4 am against 16,000 at 4 pm, so single night cells move more. Treat this as a pattern to monitor.

Footnote: fatal share = fatal collisions divided by fatal and injury collisions. Collisions with no recorded hour (0.96%) are left out of this page. Cells under 100 collisions are blank.

Data: Transport Canada, National Collision Database 2019-2020 (OGL - Canada). Fatal and injury collisions only.

Collisions from Jan 2019 to Dec 2020 | Police-reported fatal and injury collisions only

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2020 collision decline was driven almost entirely by injury collisions

2020 against 2019: the overall change, the monthly pattern, fatal against injury, and where the decline came from.

Recorded collisions fell 24.38% in 2020: injury collisions fell 24.74% and fatal collisions only 1.29%.

Reset filters

Collisions, 2020 vs 2019

105,782 in 2019 to 79,990 in 2020

-24.38%

Injury collisions, 2020 vs 2019

104,159 in 2019 to 78,388 in 2020

-24.74%

Fatal collisions, 2020 vs 2019

1,623 in 2019 to 1,602 in 2020

-1.29%

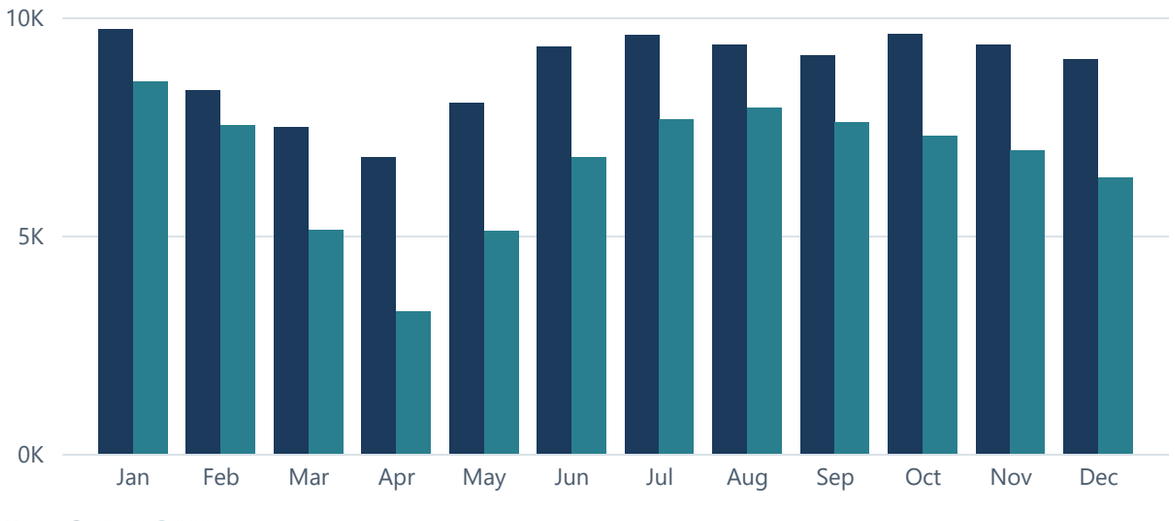
Fatal share in 2020

Up 0.47 pts from 1.53% in 2019

2.00%

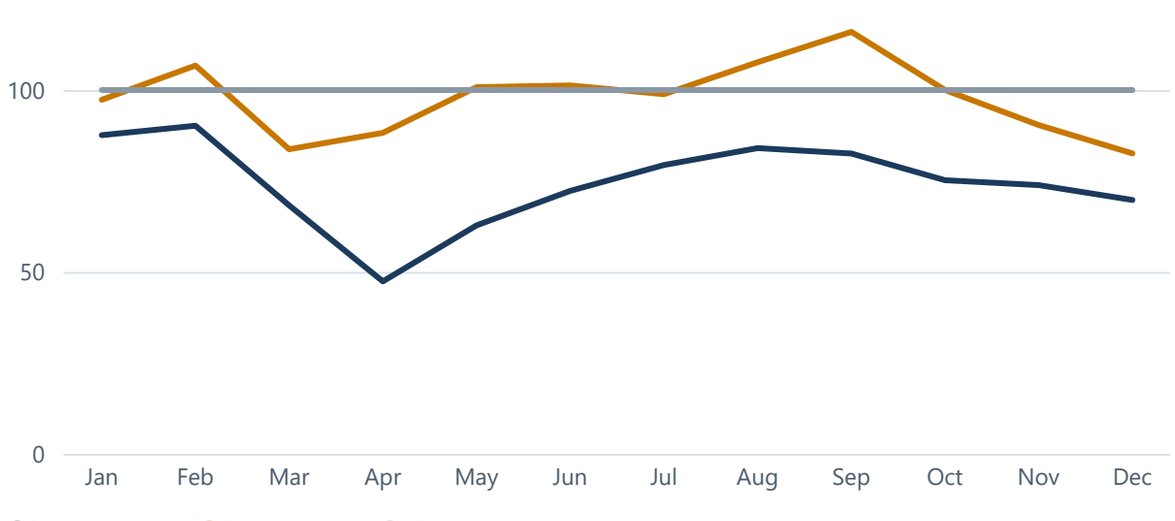
Collisions by month, 2019 against 2020

Each month has two bars. The gap in spring is the drop.



Monthly collision counts relative to 2019

100 = the same month of 2019. Below 100 means fewer recorded collisions in 2020.



Where the decline came from

Change in collisions, 2019 to 2020. Injury and fatal add up to the total decrease.

Injury collisions

99.92% of the total decrease

-25,771

Fatal collisions

0.08% of the total decrease

-21

Fatal and injury collisions: 2019 against 2020

Full-year counts, the same totals as the cards above.

Severity	2019	2020	Change	% change
Fatal	1,623	1,602	-21	-1.29%
Injury	104,159	78,388	-25,771	-24.74%
Total	**105,782**	**79,990**	**-25,792**	**-24.38%**

01 · Overall decline

-24.38%

Recorded collisions fell 24.38% in 2020. April had the largest drop: 52.10% below April 2019.

02 · Driver of the decline

99.92%

Injury collisions account for 99.92% of the decrease. Fatal collisions fell only 1.29%.

03 · Interpretation

1.53% to 2.00%

The fatal share rose because fatal collisions held while others fell. That is a change in the mix of recorded collisions, not proof that driving became riskier.

Footnote: 2020 is a pandemic year, so it is not a typical comparison. Cards and the table use all collisions in each year. Only the monthly charts leave out two collisions with no recorded month. Monthly fatal counts are small (about 70 to 200), so single months move more.

Data: Transport Canada, National Collision Database 2019-2020 (OGL - Canada). Fatal and injury collisions only.

Collisions from Jan 2019 to Dec 2020 | Police-reported fatal and injury collisions only

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What if risky roads were made as safe as safe ones?

A what-if scenario: choose how much of the gap to close and see how many fatal collisions that could avoid.

Year (none = both)

All

Closing 100% of the gap: add traffic control could avoid about 1,744 fatal collisions (54.09%).

Reset filters

Fatal collisions avoided
Lever: Add traffic control
1,744

Share of all fatal collisions
of all 3,225 fatal collisions in the period
54.09%

Share of the gap closed
Set it with the dropdown below
100%

Fatal collisions in the period
Overall fatal share is 1.74%
3,225

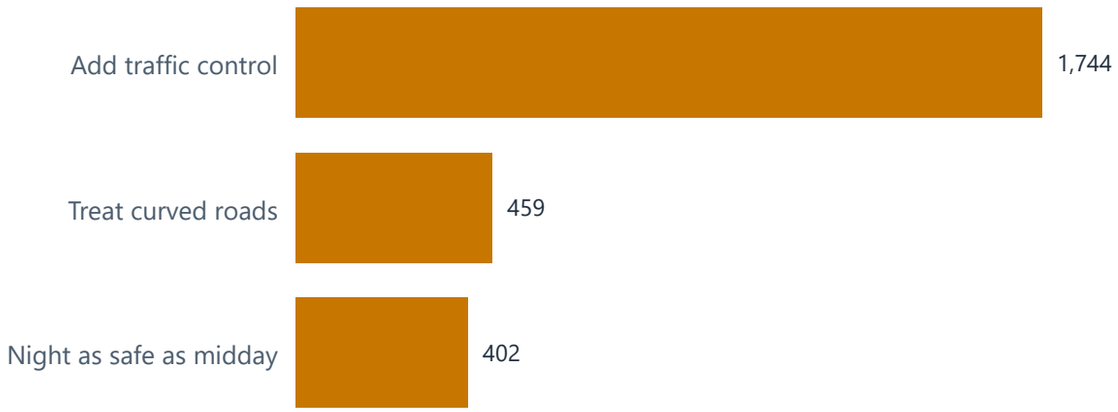
Share of the gap closed (none = 25%)

100%

How it works: for each lever, the fatal share on the risky roads is moved toward the share on the safer roads, by the percentage you choose. Avoided = fatal collisions on the risky roads x (1 - safer share / risky share) x share chosen. It is illustrative: it assumes the gap is caused by the road, which the data cannot prove.

Fatal collisions avoided, by lever

Each bar is one lever on its own. Levers overlap, so do not add them.



The numbers behind each lever

Safer share is the fatal share on signalled roads, straight roads and midday.

Lever	Fatal now	Fatal share now	Safer share	Avoided
Add traffic control	2,349	2.41%	0.62%	1,744
Night as safe as midday	701	3.41%	1.46%	402
Treat curved roads	771	3.63%	1.47%	459

01 · The biggest lever

1,744

If uncontrolled roads had the fatal share of signalled ones, about 1,744 fewer fatal collisions (54.09% of all) would have happened. That is the largest gap by far.

02 · Smaller levers

459 and 402

Treating curved roads like straight ones would avoid about 459 (14.25%), and making night as safe as midday about 402 (12.46%). These are at 100% of the gap closed.

03 · How to read this

What-if

This is a scenario, not a forecast. It uses recorded collisions with no traffic volumes, assumes the gap is caused by the road, and the levers overlap.

Footnote: fatal share = fatal collisions divided by fatal and injury collisions. Scenario results change with the Year slicer. The 25% default applies when no share is chosen.

Data: Transport Canada, National Collision Database 2019-2020 (OGL - Canada). Fatal and injury collisions only.

Collisions from Jan 2019 to Dec 2020 | Police-reported fatal and injury collisions only

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Fatality rates vary sharply by road user, safety-device use and age

Among people in reported fatal and injury collisions, motorcyclists, pedestrians and occupants with no safety device used show the highest fatality rates.

Year (none = both)

All

Occupants with no safety device used die 16.09 times as often, and motorcyclists and pedestrians are the most exposed.

Reset filters

Fatality rate: no device used
7,939 occupants, 554 deaths
6.98%

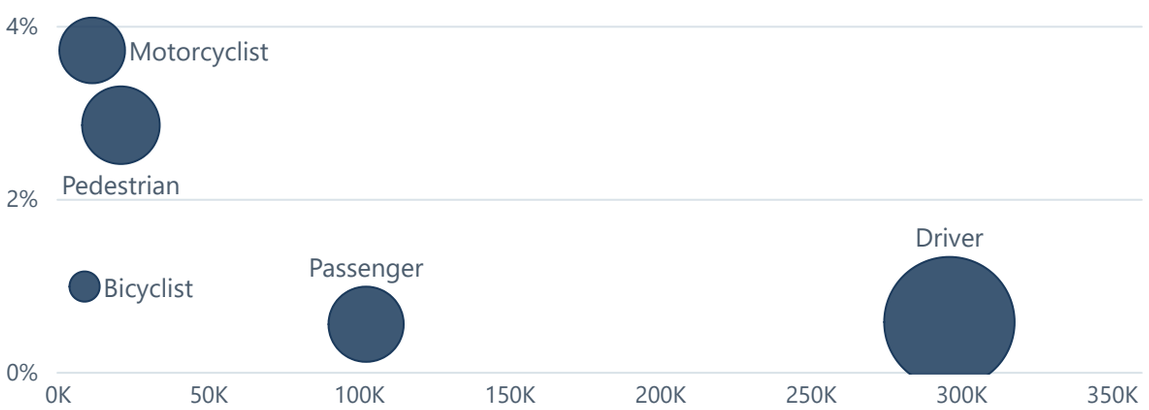
Fatality rate: device used
315,866 occupants, 1,370 deaths
0.43%

How much higher
Not adjusted for other factors
16.09×

Average fatality rate
3,508 deaths, 463,630 people
0.76%

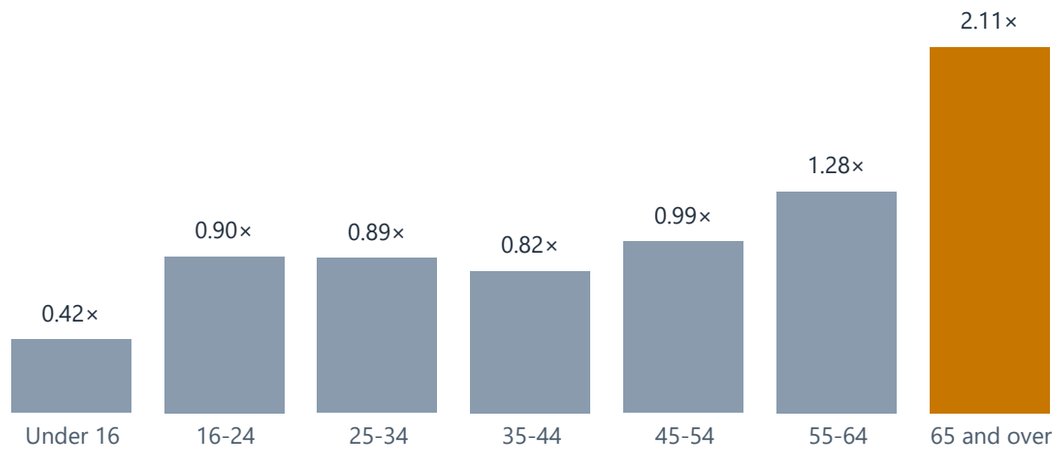
Motorcyclists and pedestrians show the highest fatality rates

Right = more people, up = higher rate, bigger bubble = more deaths.



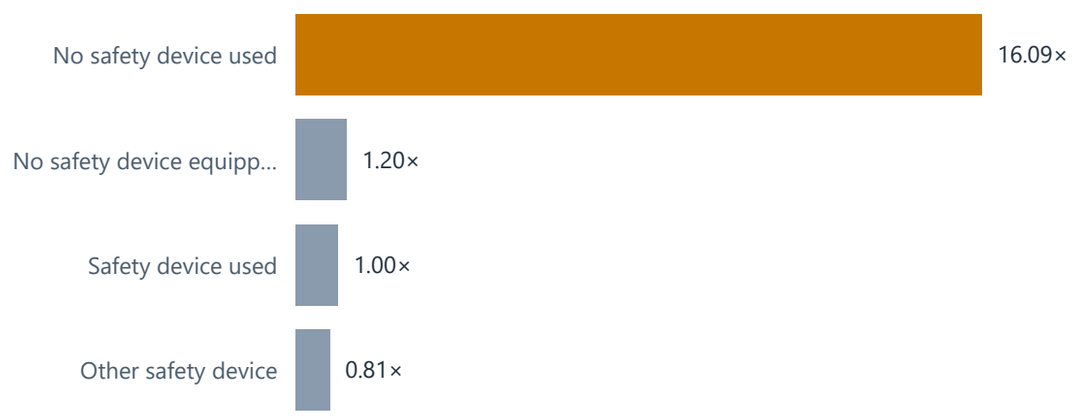
The fatality rate rises in older age groups

Times the average rate (1.00×). Amber is 1.50× or more. Unknown ages left out.



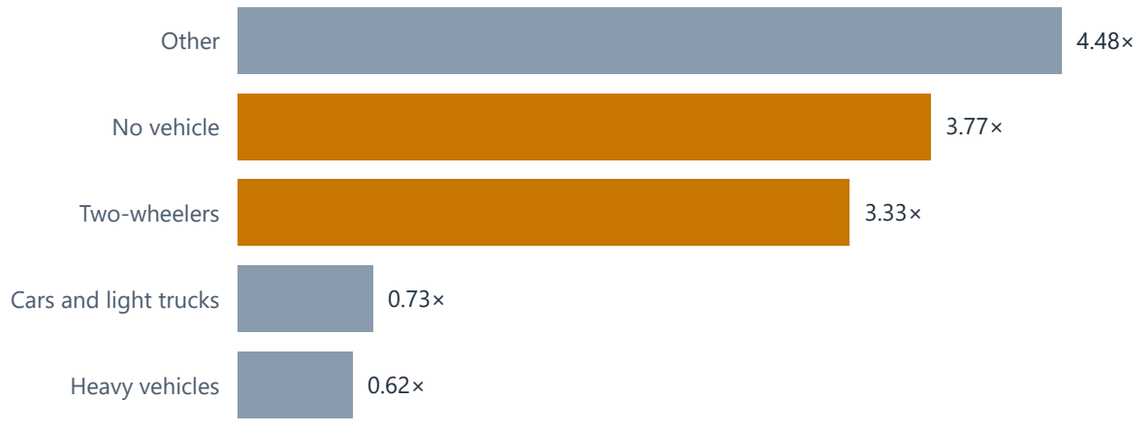
Occupants with no safety device used have a far higher fatality rate

Occupants against those who used a device (1.00×). Unknown left out.



Two-wheelers have a higher fatality rate than cars

Times the average rate. 'Other' is small (about 2,700 people): see footnote.



01 · Road users

4.91× and 3.77×

Motorcyclists show 4.91 times the average fatality rate and pedestrians 3.77 times. Drivers, the largest group, are at 0.76 times.

02 · Safety devices

16.09×

Occupants with no safety device used die 16.09 times as often (6.98% against 0.43%). The gap holds in every age band, but it is not adjusted for everything.

03 · Age

2.11×

People 65 and over show 2.11 times the average fatality rate. These are unadjusted comparisons, not the effect of age alone.

Footnote: fatality rate = deaths divided by people in reported fatal or injury collisions, not risk per trip or km. Findings use both years. 6.98%: 7,939 occupants, 554 deaths, 95% interval 6.44% to 7.56%; 0.43%: 315,866 occupants, 1,370 deaths. 'Other' vehicles: farm, construction, emergency, off-road, snowmobile, streetcar. 'Not used' differs from 'not equipped'.

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What drives a fatal outcome? Explore it yourself

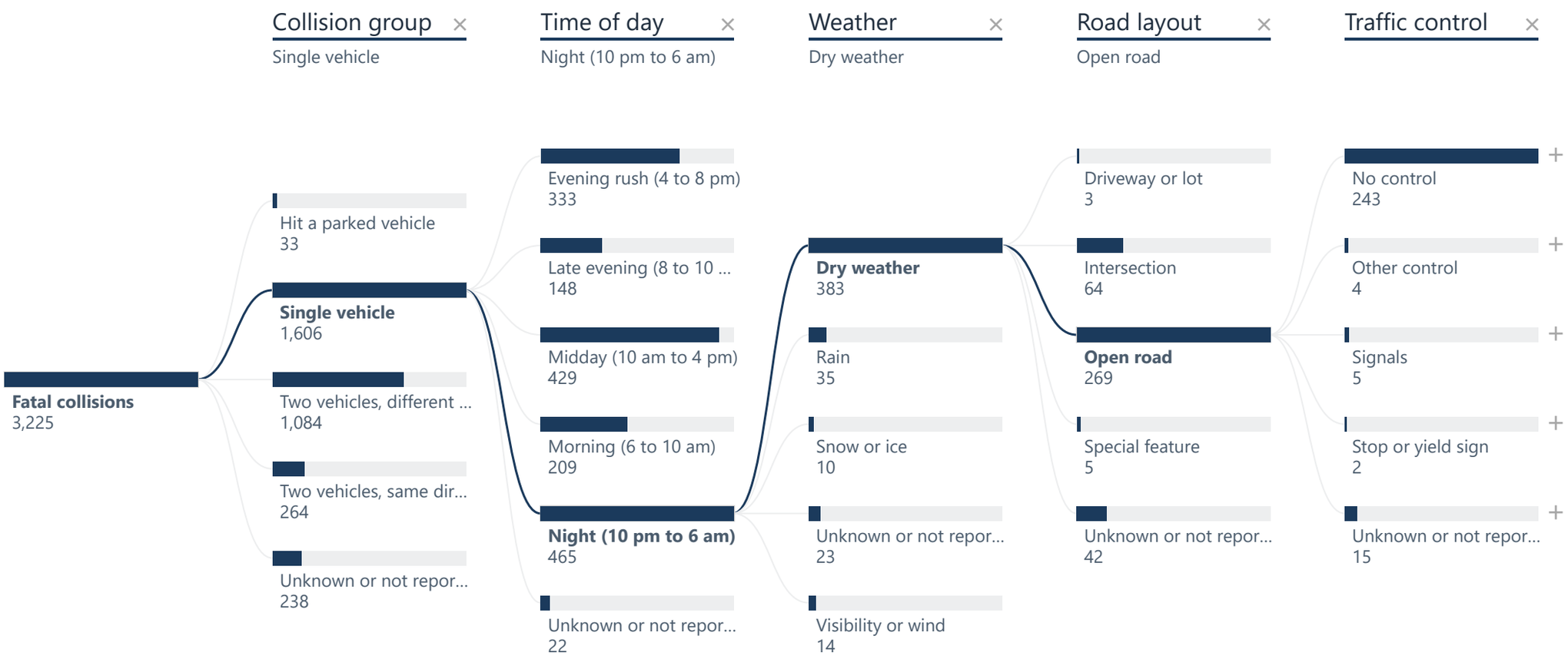
A decomposition tree: split the fatal collisions by any factor, in any order.

How to use it: click the + under a bar and choose a factor. 'High value' picks the factor that most concentrates fatal collisions, so you can see what the data finds worth splitting on.

Reset filters

Fatal collisions, split by the factors you choose

Starts with all 3,225 fatal collisions.



Footnote: the tree finds where fatal collisions concentrate by count, which favours common situations. Compare with the fatal share on the Conditions page, which shows how likely each situation is to be fatal.

Data: Transport Canada, National Collision Database 2019-2020 (OGL - Canada). Fatal and injury collisions only.

Collisions from Jan 2019 to Dec 2020 | Police-reported fatal and injury collisions only

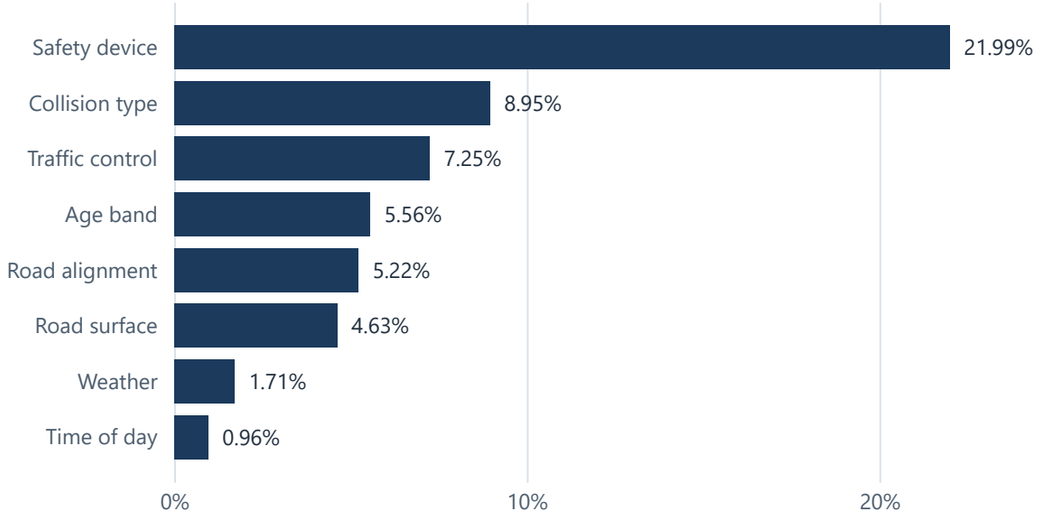
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Before any analysis: what each number in the file means

How I found the collision ID, how the model is built, and what is missing.

Share of records with an unknown value

Unknown means the police form left it blank or the jurisdiction did not provide it.



How the model is built

- A star schema: fact_collision and fact_person hold keys and counts only; 15 dimensions hold every label, group and sort order.
- Every dimension has an 'unknown' member (key 0), so records with a missing value stay in the totals and show as 'Unknown or not reported', never as (Blank).
- Dimensions have hierarchies (for example collision group to collision type) and are loaded through Power Query, which sets types and renames columns.
- One chart shows the unknown share of any field through a disconnected table and a SWITCH measure. A what-if parameter drives the Conditions scenario, and a Time Comparison calculation group (prior year, change, % change) is in the model.
- 471,046 source rows (one per person in each vehicle) include 7,416 dummy or unmatched records, such as a dummy person for a parked car. The model keeps 463,630 real people. Two collisions have no recorded month, so they appear only when no year is chosen.

How I found the collision ID

- The dictionary lists no collision ID, so my first version derived one from 'vehicle 1, person 1'. It found 185,359 collisions, 413 fewer than the real 185,772, so it had merged collisions that are separate.
- The file does hold a key, C_CASE, that the dictionary does not mention. It gives 185,772 collisions; severity, hour, weather and road surface never change inside a case, and the vehicle count matches 99.56%.
- I used C_CASE. Every fatal collision has a death and no injury collision does (3,508 deaths in 3,225 fatal collisions).

Terms in plain language

- Fatal share: the share of reported collisions that caused at least one death.
- Head-on: two vehicles hit while travelling in opposite directions.
- Fatality rate: deaths among the people in these collisions, not a risk per person on the road.
- Unknown or not reported: left blank on the police form, or not supplied by that jurisdiction.
- Only police-reported fatal and injury collisions are included, from 2019 and 2020 (the 2021 English file was blank in the catalogue).